Transfer Learning for Jet Tagging at the LHC with Pretrained Foundation Models

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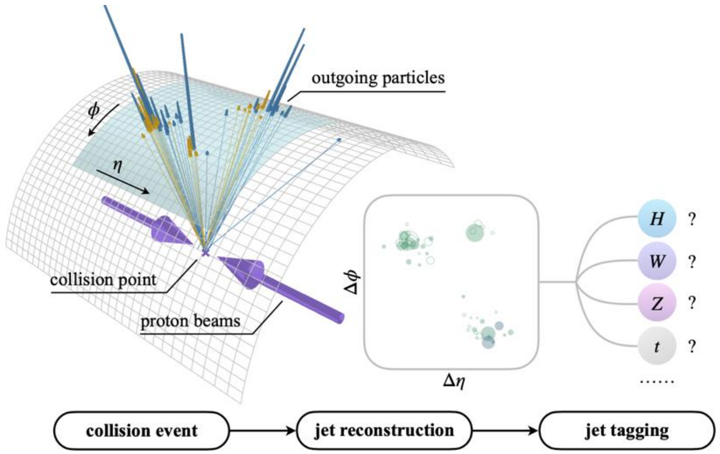
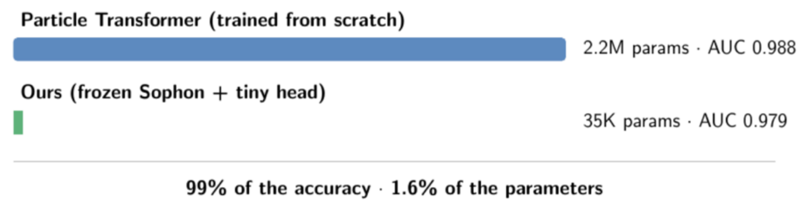
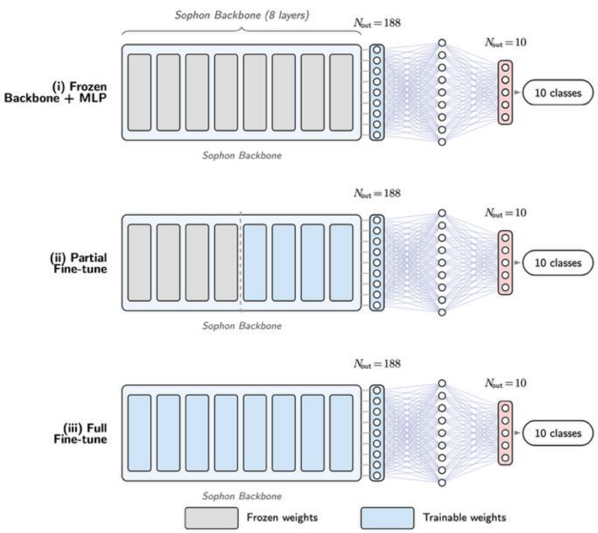
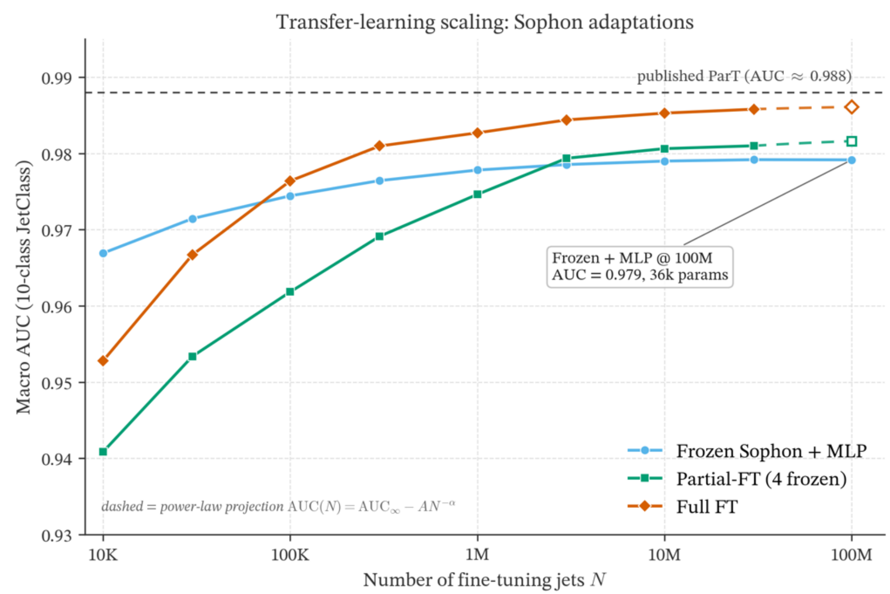


- LHC collides protons bunches ~40M times/sec, these produce jets
- Training machine learning models from scratch to identify jets is expensive







- Adapt Sophon (pretrained on 188 jet types) to a 10-class task
- 3 strategies: frozen head (35K), partial (1.4M), and full (2.2M)

- Scaling resembles resemble power-law improvement curves
- Frozen + 35K head → 99% of ParT's AUC at 1.6% of the params